CURRENCY RISK

(exchange rate risk)

Currency risk

- **Currency risk is the gains or losses that arises as a result of the movements in exchange rates.**
- **An exchange rate is the price of one currency quoted against another currency.**

Currency risk (exchange gain or loss)

- **An exchange gain or loss would occur if the exchange rate used at a time of settling the transaction is different from the rate that existed at the time of transacting.**
- **For example, a gain or loss may occur on foreign currency payment or receipt where there is an exchange rate movement between transaction date and settlement date .**

Spot exchange vs Forward rates

- A spot exchange rate is the exchange rate that is applicable for immediate transaction and settlement.
- The forward exchange rate is the rate applied for immediate transaction, but delivery and settlement are for a specific future date e.g., 1 month forward, 3 months forward etc.
- Forward exchange rates are used to lock into the price in order to reduce the currency risk on transactions.

Factors that influence exchange rates

- Political environment.
- Interest rate volatility*
- Expectation theory
- Inflation rate differential*
- Balance of payment support.

Interest rate parity (IRP):

- **The difference in interest rates between two countries can be explained by the differences in the exchange rates.**
- **Thus interest rate differential can be used to predict the future exchange rate as:**
- **Forward exchange rate = spot rate x** $\frac{(1+quoted\ currency\ interest)^n}{(1+base\ currency\ interest)^n}$

where: **Quoted currency is the variable currency.**

n is the period of time in years.

Purchasing power parity (PPP):

- **The difference in inflation rates between two countries can be explained by the differences in the exchange rates.**
- **Thus, inflation rate differential can be used to predict the future exchange rate as:**
- **Forward exchange rate = spot rate x** $\frac{(1+quoted\ currency\ inflation)^n}{(1+base\ currency\ inflation)^n}$

where: **Quoted currency is the variable currency.**

n is the period of time in years.

Activity 1

- **The spot rate for the $/K exchange is K 12.177 per $. Interest rates in New York are 3% p.a. and in Lusaka interest is 15% p.a.**
- **<u>Required:</u>**
- **Ignoring transaction costs calculate the best rate (for the customer) at which a bank will sell the kwacha.**

a) **6 months forward**

b) **12-months forward**

Solution 1

6- months forward rate =	**Spot rate * $\frac{(1+\text{quoted interest})^n}{(1+\text{base interest})^n}$** **$= 12.177 * \frac{(1.15)^{0.5}}{(1.03)^{0.5}}$ = *K12.867***
12- months forward rate =	**Spot rate * $\frac{(1+\text{quoted interest})^n}{(1+\text{base interest})^n}$** **$= 12.177 * \frac{(1.15)^{1}}{(1.03)^{1}}$ = *K13.596***

Activity 2

- **If the inflation rate in the UK is 6 per cent and in the US inflation is 4 per cent, determine the 1-year forward exchange rate Assuming an exchange rate of $1.60 = £1**

Solution 1

1- year forward rate =	**Spot rate * $\frac{(1+\text{quoted inflation})^n}{(1+\text{base inflation})^n}$** **$= 1.6 * \frac{(1.04)^1}{(1.06)^1}$ = *$1.5698***

Activity 3

- **If the inflation rate in Zambia is 15 per cent and in the US, inflation is 4 per cent, determine the 6-months forward exchange rate Assuming an exchange rate of K26.70 = US$1**

Activity 4

- **The spot exchange rate between kwacha and sterling pound is K27.035 per £1. The interest rate in the UK is 2.5% per annum, interest rate in Europe is 4% per year while the Zambian interest rate currently stands 9% per year.**
- **<u>Required</u>**
- **Calculate the 6-months and the 1-year forward rates.**

Types of currency risk

- ***Transaction risk:*** **Exchange gain/losses that arises as a result of movement in exchange rates between transaction date and settlement date.**
- ***Translation risk:*** **Exchange gain/losses that arises due to the need to consolidate company accounts that involves foreign interest (assets and or liabilities)**
- ***Economic risk:*** **loss of value or international competitiveness of a local business due to adverse movements in exchange rates over time.**

Internal hedging methods

- ***Home currency invoicing:*** **Hedger can invoice its customers in its home currency or request to be invoiced in its home currency.**
- ***Matching receipts and payments:*** **Maintaining foreign currency accounts so that receipts in one currency are matched with payments in the same currency.**
- ***Leading and lagging:*** **Speeding up collection/payments as well as delaying receipts or payments in order to take advantage of existing or anticipated favourable exchange rates.**

External hedging methods

- **Forward contracts**
- ***Futures contracts***
- ***Money market hedging***
- ***Currency options***
- ***Swaps****

Forward contract

- Is an agreement that fixes the price at which trading will take place between two parties at a specific future date e.g. 2-months forward
- Forward contract are over-the-counter (OTC) contracts i.e. are tailored to suit the needs of the parties
- Parties to the contract specifies, the size of the contract, type of currency, as well as the maturity date of the contract.
- Forward contract are brokered contract (needs to find counterparty) which can result in a perfect hedge.
- Forward contracts have default risk (check financial soundness of counterparty)

Advantages and disadvantages forward contracts

- ***Merit:***
- **Eliminates the downside risk by fixing the future price.**
- **Flexible in terms of the amount thus can result in a perfect hedge.**

- ***Demerits:***
- **Contract is legally binding on the parties (not tradable).**
- **Has default risk.**
- **No chance of benefiting from favourable price movements.**

Activity 5

- **TLC Co, whose home currency is the dollar, has recently been awarded a contract by HC Co, whose home currency is the shillings; the contract conditions state that payment will be made in shillings, six months from now. The value of the contract is 20 million shillings.**
- ***Exchange rates are quoted as:***
- **Current spot rate ($ per 1 shilling) 0.0800 0.0794**
- **Six-months forward rate 0.0781 0.0776**
- **<u>Required:</u>**
- **Determine the exchange gain/loss that will be associated with the use of the forward rate compared to the existing exchange rate.**

Solution to activity 5

- **Amount receivable based on the spot = $0.0794*20,000,000 =$1,588,000**
- **Amount receivable under fwd contract =$0.0776*20,000,000 =$1,552,000**
- **Exchange loss if fwd contract is used =$36,000**

Money market

Money markets are financial markets where short-term financial securities are traded.

For example:

1. **Treasury bills (government and the public) (*borrowing and lending*)**
2. **Commercial papers (corporations and the public) *borrowing and lending***
3. **Certificate of deposits. (banks and the public) *borrowing and lending***

Money market characteristics

- The basic idea is to avoid future exchange rate uncertainty by making the exchange at today's spot rate instead.
- This is achieved by depositing/borrowing the foreign currency until the actual commercial transaction cash flows occur.

Money market hedging

- **Hedging a for foreign currency payable.**
- **This involves creating an asset now (depositing) in foreign currency so that the amount payable is covered by proceeds from the deposit.**
- **Deposit the present value of the foreign currency payable.**

Hedging a foreign currency payable

Future foreign cash flows?	• Payment?
Now	• (Borrow in domestic currency) • Buy the present value of the future payment today at spot • Place on deposit (earn interest)
Future transaction date	• Use deposit to pay supplier • (Value of deposit = accounts payable)

Money market hedging (for foreign currency payable):

- **Steps involved are:**
- **Amount to deposit = $\frac{\text{Foreign currency payable}}{1 + \text{foreign currency interest}}$**
- **Using the spot exchange rate, convert the amount to deposit to local currency for you to determine the cost of raising this amount.**
- **Determine the final cash flows based on the assumption that the local currency is borrowed:**
- **Final cash flows = Amount borrowed*(1 + local currency interest rate)**
- **NB: Interest is assumed to be simple interest.**

Money market hedging

- **Hedging a for foreign currency receivable.**
- **This involves <u>creating a liability now</u> (borrowing) in foreign currency so that the amount receivable covers the loan.**
- **Borrow the present value of the foreign currency receivable.**

Hedging a foreign currency receivables

Future foreign cash flows?	• Receipt?
Now	• Borrow the present value of the future receipt • Convert today at the spot rate. • (Place domestic currency on deposit)
Future transaction date	• Settle liability with receipt from customer • (Value of liability = accounts receivable)

Money market hedging (for foreign currency receivable)

Steps involved are:

- Amount to borrow = $\frac{\text{Foreign currency receivable}}{1 + \text{foreign currency interest}}$
- Using the spot rate, convert the amount to borrow to local currency.
- The local currency is then deposited for the receivables period.
- Final cash flows = deposit amount *(1 + Local currency interest rate)

Activity 6

HT Co, a UK company, is in the process of negotiating a sale worth $1.4m with a large US retailer on six months' credit. The company's treasury department has provided the following:

The current spot rate is	**$1.8960 – $1.8970 = £1**
The six-month forward rate is	**$1.8840 – $1.8860 = £1**

Interest rate:	**UK**	**USA**
Annual borrowing rate	**4%**	**8%**
Annual deposit rate	**3%**	**6%**

Required:

- **Determine how much HT would receive in £ in six months if they used a money market hedge?**
- **Calculate (to the nearest £) the sterling receipt for HT if they invoiced in dollars and used a forward exchange contract.**
- **If the exchange rate were to move to $1.8870 – $1.8900 = £1 in six months' time, what would be the exchange gain or loss compared to today if the company did not hedge the transaction?**

Solution to Activity 6 (a)

- **Part A:**
- **Amount receivable in six months $1,400,000**
- **Amount to borrow = foreign currency receivable / 1 + interest**
- **=$1,400,000 / 1.04** **Interest for borrowing = 8%pa or 4%/6 months**
- **=$1,346,154**
- **Convert the dollar loan to sterling at spot i.e., $1.8970 to £1**
- **Amount of sterling = $1,346,154/$1.8970**
- **= £709,622**
- **Deposit the sterling (local currency) for the receivables period.**
- **Final cash flows = deposit * (1+ interest)** **Interest for deposit =3%pa or 1.5%/6months**
- **= £709,622*1.015**
- **= £720,267**

Solution to Activity 6 (b)

- **Part B:**
- **Calculate (to the nearest £) the sterling receipt for HT if they invoiced in dollars and used a forward exchange contract.**
- **Amount receivable = $1,400,000**
- **Six months forward exchange rate = $1.8860 per £1**
- **Sterling receipt = $1,400,000/ $1.8860**
- **= £742,312**

Solution to Activity 6 (b)

- **Part C:**
- **If the exchange rate were to move to $1.8870 – $1.8900 = £1 in six months' time, what would be the exchange gain or loss compared to today if the company did not hedge the transaction?**
- **Amount receivable = $1,400,000**
- **Based on the spot exchange, amount receivable = $1,346,154/$1.8970 = £709,622**
- **Value in six months =720,267**
- **Based on expected spot rates in six months = $1,400,000/$1.8900= £740,741**

Solution to Activity 6 (b)

- **<u>Alternatively, Part C :</u>**
- **If the exchange rate were to move to $1.8870 – $1.8900 = £1 in six months' time, what would be the exchange gain or loss compared to today if the company did not hedge the transaction?**
- **Assuming the full of accounts receivable = $1,400,000 is converted at spot**
- **= $1,400,000/$1.8970 = <u>£738,007</u>**
- **Based on expected spot rates in six months = $1,400,000/$1.8900= <u>£740,741</u>**

Futures contract

- Is an *exchange traded* agreement that fixes the price at which an underlying asset will be traded future.
- Futures contracts are standardized contracts in terms of contract size, type of currency, and settlement dates (maturity date).
- These requires an initial margin and marking-to-market.
- Contracts can be closed out by entering into a second contract with the same value or deliver the underlying asset at maturity (rare case).
- The ability to close-out on a contract makes futures contracts to be highly attractive but they require sound financial management knowledge.

Advantages and disadvantages futures contracts

- ***Merit:***
- **Has lower transaction costs compared to other hedging methods.**
- **There is price transparency because futures are tradable contracts.**
- ***Demerits:***
- **Contract cannot be tailored to the needs of the parties.**
- **Limited to specific currencies.**
- **Results in hedge inefficiencies due to standardization.**

- **Overall, the gain on the futures position will be offset by the loss on the spot price and vice versa.**

Options

- An option is contract that gives the option holder(buyer) the right but not the obligation to buy/sell the underlying asset at some point in future at a predetermined price.
- Option is exercised when the contract price is favorable (option is in-the-money).
- Option is allowed to lapse when the price is adverse (out-of-the-money).
- Option is at-the-money when contract price = current security price
- Price (value) of an option is the premium paid to the option writer (seller).
- The flexibility of an option lies with the option holder and not the writer.

Options cont.

- A Put option is an option to sell the underlying asset.
- A Call option is an option to buy the underlying asset
- Options can OTC or exchange traded
- European options can only be exercised at expiry.
- American options can be exercised any time up to time to expiry
- Options are expensive because of the premium that has to be paid upfront to the option writer.
- A covered call is a call option written by the owner of the asset.
- A naked call is written by one that does not own the underlying asset.

Advantages and disadvantages options

- ***Merit:***
- **Options holder takes advantage of favorable price movements.**
- **OTC options are tailor-made and can be arranged for longer periods.**
- **Traded options = price transparency, better regulation and liquidity.**
- ***.Demerits:***
- **Requires an upfront payment (premium).**
- **Traded options are not available in every currency.**
- **Cost (premium) depends on the expected price volatility (risk).**